COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.5

This worksheet contains a combination of *formative activities* (which contribute towards your learning) and *summative activities* (which you will complete and submit to be assessed as part of your portfolio).

- Exercises marked with red borders are summative and must be submitted in Gradescope.
- Other exercises may be covered in tutorial sessions.
- Exercises marked by (*) are advanced, and may take some time to complete.
- Exercises marked by (*E*) are exam-style questions. You should expect to see questions like this in the exam.

Expectations:

1. **Timeliness** You should complete all of the activities in the order provided and submit your portfolio evidence on PebblePad before the deadline.
2. **Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff.
3. **Integrity:** You are responsible for ensuring that the evidence you submit as part of your portfolio is **entirely your own work**. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in tutorials and the Math Lunch Hour. Individual support is available via office hours.

Expected completion date: Friday 6 November 2026

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment. Within this assessment you may use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries.
- test your knowledge against a piece of content.
- help identify and correct spelling mistakes and grammatical errors in your work.
- provide feedback and advice on your overall coding style.

You must not use Gen AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning outcomes

On completion of this activity sheet you will have:

1. explained and reasoned about the concept of cardinality,
2. solved problems by applying counting principles, such as the addition and multiplication principles,
3. applied the binomial theorem to problems arising in computer science,
4. applied the pigeonhole principle to solve problems arising in computer science,
5. applied recursive definitions of sequences or structures.

Tutorials on Wednesday and Thursday will cover material from Week 5, Lecture 1. This material is in exercises 1–8. Tutorials on Monday and Tuesday will cover material from Week 5, Lecture 2. This material is in exercises 9–14.

Questions

1. Car number plates in the UK have the format

   Letter Letter Digit Digit Letter Letter Letter

   (Assume we have 26 letters and 10 digits.) In New York state, number plates have the format

   Letter Letter Letter Digit Digit Digit Digit

   Are there more possible number plates in the UK, or in New York?

2. A standard pack of cards has 52 cards. They are divided into four *suits*: the hearts, diamonds, clubs, and spades. There are 13 cards in each suit, labelled $2, 3, 4, 5, 6, 7, 8, 9, 10$, Jack, Queen, King, and Ace. This label is called the *rank* of the card. A *hand* is a set of five cards chosen from the pack.

   (a) How many ways are there of choosing a hand?

   (b) A *full house* consists of three cards with the same rank along with two cards that have the same rank (and this rank is different from the rank of the first three cards). How many hands are full houses?

   (c) A *four of a kind* consists of four cards with the same rank, along with one other card. How many hands are four of a kind?

   (d) If a hand is selected at random, is it more likely to be a full house or a four of a kind?

3. (*E*) How many binary strings with 20 digits contain 12 copies of 1 and 8 copies of 0?

4. (*E*) We are going to make a string of 10 digits, where each digit is in $\{0, 1, 2\}$. Each of the digits can be any one of 0, 1, or 2. We require that the string contains exactly two copies of 0, three copies of 1, and five copies of 2. We want to find the number of these strings.

   (a) Explain why the answer is equal to

   $$\binom{10}{2}\binom{8}{3}$$

(b) Explain why the answer is also equal to

$$\binom{10}{5}\binom{5}{2}$$

5. Let m be a positive integer. Use the binomial theorem to prove the following facts.

   (a)

   $$\sum_{i=0}^{m} \binom{m}{i} 2^i = 3^m$$

   (b)

   $$\sum_{i=0}^{m} \binom{m}{i} = 2^m$$

   Can you explain the last fact without using the binomial theorem?

6. (*) Let m be a positive integer and let r be a non-negative integer. Use induction on r to prove that

   $$\binom{m}{0} + \binom{m+1}{1} + \binom{m+2}{2} + \cdots + \binom{m+r}{r} = \binom{m+r+1}{r}.$$

   Can you see why this is sometimes called the *hockey-stick identity*?

7. A standard pack of cards has 52 cards. They are divided into four *suits*: the hearts, diamonds, clubs, and spades. There are 13 cards in each suit, labelled $2, 3, 4, 5, 6, 7, 8, 9, 10$, Jack, Queen, King, and Ace. This label is called the *rank* of the card. A *hand* is a set of five cards chosen from the pack.
   A hand of cards is a *two-pair* if it contains two cards which have the same rank as each other, another two cards which have the same rank as each other (but a different rank from the first pair), and a fifth card which has a different rank from all the others.
   How many hands are two-pairs? Explain your answer.

8. (*E*) Let A be a set with cardinality 20 and let B be a set with cardinality 6. Assume f is a function from A to B. Explain why there is an element $b \in B$ such that $|\{a \in A \mid f(a) = b\}| \geq 4$.

9. Let A be the set $\{1, 2, 3, 4, 5, 7, 8, 12, 14, 15, 24\}$. Let R be the relation on A defined so that aRb is true if and only if a divides b. Draw a Hasse diagram for R.

10. (*E*) In a class of 100 students every student has at least one of the subjects computer science, mathematics, or physics as a major. There are 65 with a computer science major, 46 with a mathematics major, and 37 with a physics major. There are also 21 studying computer science and maths, 12 studying computer science and physics, and 22 studying maths and physics. How many student are have all three subjects majors?

11. We want to count the number of strings of 12 characters made up of the symbols A, B, and C, where each symbol has to appear at least once.

    (a) How many strings of length 12 can we make from the symbols A, B, and C (without any constraints)?

    (b) Let S_A be the set of strings which *do not* contain any copies of A. Define S_B and S_C in the same way. What are the cardinalities of S_A, S_B, and S_C?

(c) What are the cardinalities of $S_A \cap S_B$, $S_A \cap S_C$, and $S_B \cap S_C$?

(d) What is the cardinality of $S_A \cap S_B \cap S_C$?

(e) How many strings contain at least one copy of A, B, and C?

12. How many ways are there of ordering the letters in the English alphabet if we require that the words *fox* and *quick* all appear in consecutive letters?

13. How many ways are there of ordering the 26 letters in the (English) alphabet if we require that the words *am*, *dog*, *pink*, and *flute* all appear in consecutive letters. For example, we want to count orderings like the following one.

dogzsflutewxjqpinkhbramyvc

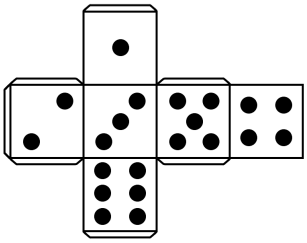
You should explain your answer.

14. Let A be a set satisfying $|A| = 6$. Let B be the set $\{x, y\}$. Explain why the number of onto functions from A to be B is $2^6 - 2$.

15. (*) How many ways are there of seating n people around a circular table? We consider two seatings to be the same if one can be obtained from the other by asking everyone to stand up and move some constant number of positions to their left. (Another way of saying the same thing: we consider two seating to be equivalent if they are the same up to rotation. How many equivalence classes are there?)

16. (*) A *Rubik's cube* has six faces, and each face is divided into 3×3 grid of squares. The cube is constructed from a number of pieces. Each piece has at least one coloured square on it. (There is a mechanism in the middle which holds things together, but we will not count this as a piece: we only want to count the pieces which have a coloured square.) We will use inclusion-exclusion to count the number of pieces.





Label the faces of the cube 1, 2, 3, 4, 5, 6 in the same way as the faces of a six-sided die (see the diagram). Note that 1 and 6 are opposite faces, and so are 3 and 4, and 2 and 5. When i is in $\{1, 2, 3, 4, 5, 6\}$, let S_i be the set of pieces that appear on face i.

(a) What is the number

$$\sum_{\substack{I \subseteq \{1,2,3,4,5,6\} \\ |I|=1}} \left| \bigcap_{i \in I} S_i \right|$$

(b) What is the number

$$\sum_{\substack{I \subseteq \{1,2,3,4,5,6\} \\ |I|=2}} \left| \bigcap_{i \in I} S_i \right|$$

(c) What is the number

$$\sum_{\substack{I\subseteq\{1,2,3,4,5,6\}\\|I|=3}}\left|\bigcap_{i\in I}S_i\right|$$

(d) When $I\subseteq\{1,2,3,4,5,6\}$ and $|I|\geq 4$, then

$$\bigcap_{i\in I}S_i$$

is empty. Why is this?

(e) Use inclusion-exclusion to find the number of pieces in a Rubik's cube.

(f) Can you find an alternative explanation for the answer in (e)?

17. (*) A *derangement* of the numbers $1,2,3,\ldots,n$ is an ordering such that there does not exist a number i that is in position i. For example, 2341 is a derangement of $1,2,3,4$, since no number is in its own position. On the other hand, 3124 is not a derangement, since number 4 is in position 4.

Let S_i be the set of orderings that have number i in position i.

(a) What is the total number of orderings of $1,2,\ldots,n$?

(b) What is the total $|S_1|+|S_2|+\cdots+|S_n|$?

(c) Subtract the answer to (b) from the answer in (a).

(d) What is the total

$$\sum_{\substack{I\subseteq\{1,2,\ldots,n\}\\|I|=2}}\left|\bigcap_{i\in I}S_i\right|$$

(e) Add the answer from (d) to the answer from (c).

(f) Continuing in this way, and using inclusion-exclusion, can you find a formula for the number of derangements of $1,2,\ldots,n$?

(g) What is the probability that a randomly selected ordering will be a derangement? What happens to this probability as n gets larger and larger?